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AIML MATTERS

Newsletter

Dec 2024- Jan 2025

Department Of Artificial Intelligence & Machine learning

Issue 16

Frequency : Monthly

Email : hod_aiml@jnnce.ac.in

AIML FLASH NEWS

- Preface -

A recent study published in PLOS ONE reveals that people tend to assign greater blame to artificial intelligence for moral transgressions when they perceive it as having human-like qualities, such as awareness, thinking, and decision-making abilities. Researchers found that in cases where AI was involved in real-world ethical failures—such as biased decision-making or harmful autonomous actions—participants attributed more blame to the AI itself and less to human stakeholders, including developers, companies, or governing bodies. These findings highlight concerns about AI being used as a moral scapegoat, potentially shifting responsibility away from those who design and deploy it.

As real human-generated data becomes scarce, tech companies are increasingly relying on synthetic data to train AI models. While synthetic data offers advantages like scalability, cost-effectiveness, and privacy protection, over-reliance on it can lead to AI degradation, hallucinations, and loss of accuracy. Experts emphasize the need for global standards, human oversight, and validation systems to ensure AI remains reliable, transparent, and effective.

"A year spent in Artificial Intelligence is enough for one to believe in God"
- Alan Perlis

FACULTY DESK

We are delighted to welcome Ms. Aaliya Waseem as an Assistant Professor in the AI & ML Department. A dedicated academician with a strong foundation in machine learning and cloud computing, Ms. Aaliya brings a wealth of knowledge and passion for teaching to our institution.

Hailing from Shivamogga, she has pursued her entire education in her hometown, earning a BE in Information Science and Engineering from PESITM, Shivamogga. After gaining industry experience as a freelance content writer for Entrepreneur First, a Delhi-based magazine, she further honed her expertise by pursuing an M.Tech at JNNCE, where she secured the VTU 1st rank and a gold medal in 2024. Her research interests lie in edge computing, reflecting her deep engagement with emerging technologies.

Ms. Aaliya's journey into academia is fueled by a lifelong passion for teaching, transforming her childhood fascination with education into a fulfilling career. She is committed to creating an engaging and inspiring learning environment for students, helping them navigate the dynamic world of AI & ML.

We look forward to her contributions and wish her a successful journey with us!

Dr. Chetan K R took on the prestigious role of **Mentor for the International Collegiate Programming Contest** (**ICPC Asia**). Throughout **December 2024**, he guided and trained students to enhance their competitive programming skills, preparing them for one of the most challenging coding competitions in the world.

FACULTY DESK

Mrs. Shaziya Banu S successfully completed the **Fundamentals of Deep Learning** certification from **Nvidia** on **December 1, 2024**. This certification deepened her expertise in advanced deep learning concepts, equipping her with the latest skills to integrate AI-driven solutions into academic and research projects.

Mr. Ranjan V and Dr. Manu A P made a significant research contribution by publishing their paper titled "A Survey on Outlier Detection in Videos Using Machine Learning Methods: Embracing New Trends and Technologies" in the GIS-Zeitschrift für Geoinformatik GIS Science Journal.

- Publication Details: UGC Indexed, Volume 11, Issue 12, Pages 284-292
- Publication Date: December 16, 2024
- Research Focus: This survey explores the advancements in outlier detection techniques in videos, leveraging modern machine learning methods to enhance accuracy and efficiency in anomaly detection.

Dr. Ashwini J P, Girish Mantha, and Poornalatha G contributed to the research community by presenting their paper "Exploration Approaches for Identifying Key Mineral Resource Prospects" at the Fourth IEEE International Conference on Multimedia Processing, Communications, and Information Technology (MPCIT-2024).

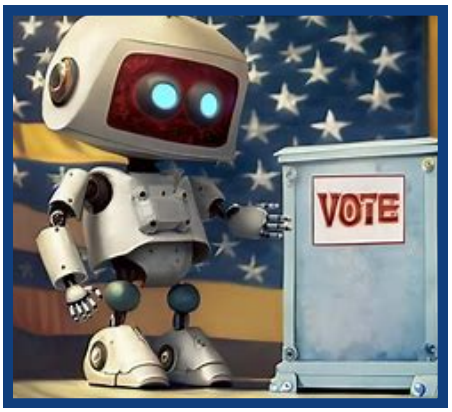
Event Details: Hosted at JNNCE, Shivamogga, on December 13-14, 2024.

DOI: 10.1109/MPCIT62449.2024

Research Impact: Their work provides innovative approaches to identifying critical mineral resource sites, integrating AI, geospatial analysis, and predictive modeling to enhance exploration strategies.

BLOGS AND ARTICLES

LLMs in Cybersecurity



Threat intelligence is a vital component of cybersecurity, helping analysts detect and respond to cyber threats. However, the sheer volume of data, unstructured sources, and constantly evolving attack techniques make manual analysis difficult. Large Language Models (LLMs) are transforming this field by automating threat intelligence tasks, such as summarizing reports detecting threats, and profiling cybercriminal groups.

They enhance efficiency by extracting key indicators of compromise (IOCs), identifying attack patterns, and providing real-time insights from various intelligence sources, including social media and security blogs. LLMs utilize advanced techniques like prompt engineering, few-shot learning, and Retrieval-Augmented Generation (RAG) to improve accuracy and relevance in threat analysis. They enable security professionals to interact with AI using natural language queries, making complex cybersecurity data more accessible. Real-world applications, such as Microsoft Security Copilot and The Intel Brief by SecurityBreak, showcase how LLMs assist in automated intelligence gathering and decision-making. By leveraging AI-driven security tools, organizations can proactively identify risks and enhance their cybersecurity posture. Despite their advantages, LLMs also pose challenges, including false positives, data privacy concerns, and potential misuse by cybercriminals. Ethical considerations and rigorous validation are essential to ensure reliable and secure applications of AI in cybersecurity. As threats continue to evolve, integrating LLMs into threat intelligence will be crucial in staying ahead of attackers, making cybersecurity operations more efficient and proactive.

-KRUTHI B R
6th SEMESTER

FOODATHON

On December 5, 2024, the Department of Artificial Intelligence and Machine Learning conducted FOODATHON 2024 – Flame-Free Feast at AIML Classrooms. The event encouraged students to showcase their culinary creativity by preparing dishes without the use of fire while promoting healthy and innovative food choices. With enthusiastic participation, teams competed by crafting unique and nutritious dishes using fresh ingredients, adhering strictly to the flame-free rule. The competition aimed to foster teamwork, creativity, and awareness of healthy eating habits, participants presented their dishes with detailed explanations of their ingredients, preparation methods, and nutritional benefits. The event provided a platform for students to experiment with alternative cooking techniques, including blending, mixing, steaming, and refrigeration to create visually appealing and delicious dishes. Judges evaluated the entries based on creativity, presentation, taste, and nutritional value, making the competition both exciting and educational. In addition to the competition,

the event also featured interactive discussions on mindful eating sustainable food practices, and the importance of incorporating fresh, wholesome ingredients into daily meals. The enthusiasm and innovation displayed by the participants highlighted their dedication to healthy living and culinary artistry.

CANVA WORKSHOP

1st Place

Anagha C, Dhanyashree Urs, Prajna N Kulal, Yashaswini S Gowda

2nd Place

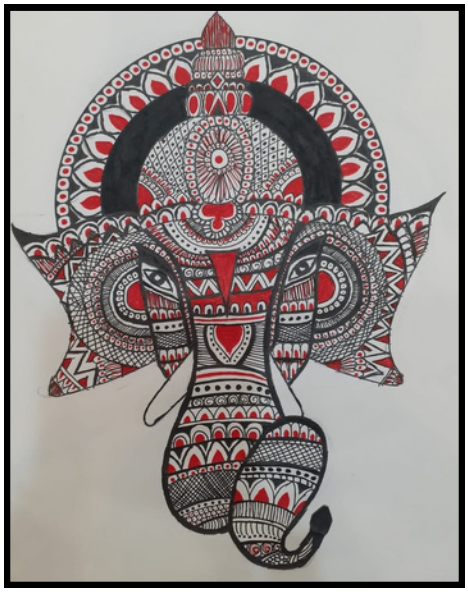
Sahana N Devadiga, Prajwal S

3rd Place

Devika N D, Deekshitha T M, Komal, Shreya R Chittaragi

The event concluded on a high note, leaving participants inspired and motivated to explore more flame-free cooking techniques while maintaining a healthy lifestyle. The success of FOODATHON 2024 demonstrated the perfect blend of creativity, nutrition, and teamwork, making it a memorable experience for all involved.

"STUDENTS' ARTWORKS"



VIDYASHREE Y C
6th Semester

SINCHANA K
4th Semester

NAGASHREE S P
6th Semester

DEVIKA N D
4th Semester

RIDDLES

1

I learn from data, improve with experience, but I never go to school. What am I?

Answer: A Machine Learning Model

2

I can recognize faces, but I don't have eyes. I can predict outcomes, but I don't have a brain. What am I?

Answer: An AI Algorithm

3

The more data I get, the smarter I become, but if you give me too much, I might just fail. What am I?

Answer: A Neural Network

4

The more data I get, the smarter I become, but if you give me too much, I might just fail. What am I?

Answer: A Neural Network

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